



Red Hat OpenShift AI Self-Managed 3.4

Working with model registries

Working with model registries in Red Hat OpenShift AI Self-Managed

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Working with model registries in Red Hat OpenShift AI Self-Managed

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Abstract

As a data scientist or AI engineer in OpenShift AI, you can register, store, version, deploy, and track models by using a model registry.

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PREFACE

As a data scientist or AI engineer in OpenShift AI, you can register, store, version, deploy, and track models by using a model registry.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog that includes models from different source providers. OpenShift AI administrators can manage and govern model catalog sources to control which models are available to data scientists and AI engineers. For more information, see [Manage and govern model catalog sources](#).

For more information about how data scientists and AI engineers discover and evaluate models in the catalog, see [Working with the model catalog](#).

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see [Managing model registries](#).

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see [Working with model registries](#).

CHAPTER 2. WORKING WITH MODEL REGISTRIES

2.1. REGISTERING A MODEL FROM THE DASHBOARD

As a data scientist or AI engineer, you can register a model from the OpenShift AI dashboard and create the first version of the new model.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- You have access to an available model registry in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to register a model in.
3. Click **Register model**.
The **Register model** dialog opens.
4. In the **Model details** section, configure details to apply to all versions of the model:
 - a. In the **Model name** field, enter a name for the model.
 - b. Optional: In the **Model description** field, enter a description for the model.
5. In the **Version details** section, enter details to apply to the first version of the model:
 - a. In the **Version name** field, enter a name for the model version.
 - b. Optional: In the **Version description** field, enter a description for the first version of the model.
 - c. In the **Source model format** field, enter the name of the model format, for example, **ONNX**.
 - d. In the **Source model format version** field, enter the version of the model format.
6. In the **Model location** section, specify the location of the model by providing either object storage details, or a URI.
 - a. To provide object storage details, ensure that the **Object storage** radio button is selected.
 - i. To autofill the details of an existing connection:
 - A. Click **Autofill from connection**
 - B. In the **Autofill from connection** dialog that opens, from the **Project** drop-down list, select the project that contains the connection.
 - C. From the **Connection name** drop-down list, select the connection that you want to use.
This list contains only object storage types which contain a bucket.

- D. Click **Autofill**.
- ii. Alternatively, manually fill out your object storage details:
 - A. In the **Endpoint** field, enter the endpoint of your S3-compatible object storage bucket.
 - B. In the **Bucket** field, enter the name of your S3-compatible object storage bucket.
 - C. In the **Region** field, enter the region of your S3-compatible object storage account.
 - D. In the **Path** field, enter a path to a model or folder. This path cannot point to a root folder.
- b. To provide a URI, ensure that the **URI** radio button is selected.
 - i. In the **URI** field, enter the URI for the model.



IMPORTANT

Deployment of models that are registered by using a URI is currently supported for public OCI repositories only.

7. Click **Register model**.

Verification

- The new model and version details are displayed on the **Details** tab for the model version.
- The new model and version are displayed on the **Model registry** page.

2.2. REGISTERING A MODEL VERSION

You can register a new model version.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- You have access to an available model registry in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to register a model version in.
3. In the **Model name** column, click the name of the model that you want to register a new version of.
The details page for the model opens.
4. Click the **Versions** tab, and then click **Register new version**.
5. In the **Version details** section, enter details to apply to this version of the model:

- a. In the **Version name** field, enter a name for the model version.
 - b. Optional: In the **Version description** field, enter a description for this version of the model.
 - c. In the **Source model format** field, enter the name of the model format, for example, **ONNX**.
 - d. In the **Source model format version** field, enter the version of the model format.
6. In the **Model location** section, specify the location of the model by providing either object storage details, or a URI.
- a. To provide object storage details, ensure that the **Object storage** radio button is selected.
 - i. To autofill the details of an existing connection:
 - A. Click **Autofill from connection**.
 - B. In the **Autofill from connection** dialog that opens, from the **Project** drop-down list, select the project that contains the connection.
 - C. From the **Connection name** drop-down list, select the connection that you want to use.
This list contains only object storage types which contain a bucket.
 - D. Click **Autofill**.
 - ii. Alternatively, manually fill out your object storage details:
 - A. In the **Endpoint** field, enter the endpoint of your S3-compatible object storage bucket.
 - B. In the **Bucket** field, enter the name of your S3-compatible object storage bucket.
 - C. In the **Region** field, enter the region of your S3-compatible object storage account.
 - D. In the **Path** field, enter a path to a model or folder. This path cannot point to a root folder.
 - b. To provide a URI, ensure that the **URI** radio button is selected.
 - i. In the **URI** field, enter the URI for the model.



IMPORTANT

Deployment of models that are registered by using a URI is currently supported for public OCI repositories only.

7. Click **Register new version**.

Verification

- The new model version is displayed in the **Latest versions** section on the **Overview** tab on the model details page.
- The new model version is displayed in the **Latest version** column on the **Model registry** page.

2.3. REGISTERING AND STORING A MODEL AS AN OCI IMAGE

You can register and store a model as an Open Container Initiative (OCI) image from the OpenShift AI dashboard. You can register a model from object storage or a URL and transform it into an OCI ModelCar image.

The ModelCar target format enables fast deployment with KServe. The model transfer job runs as a background Kubernetes Job that you can also monitor from the dashboard.

Prerequisites

- You have logged in to the Red Hat OpenShift AI dashboard.
- You have access to an available model registry in your deployment.
- If using object storage as the model origin, you have the required credentials.
- You have credentials for the OCI registry in which to store the ModelCar image.

Procedure

1. In the OpenShift AI dashboard, click **AI hub** → **Models** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry for the model.
3. Click **Register model** to create a new model, or navigate to an existing model, and click **Register new version**.



NOTE

You can also start from the model catalog and register a model in the catalog into your registry. For more details, see [Working with the model catalog](#).

4. In the **Model location and storage** section, select **Register and store**.
The **Register and store** option starts an async Kubernetes Job that copies the model into an OCI registry as a ModelCar-format image and then registers the metadata.

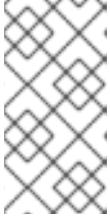
The **Register** option registers metadata only and points to an artifact that is already stored elsewhere.



NOTE

If you register a catalog model whose source is already an **oci://** URI, the UI forces **Register** mode only because the model is already stored in OCI format.

5. In the **Model transfer job name** field, provide a human-readable display name for the job, for example, **Upload granite-7b to Quay**.
6. Optional: Click **Edit resource name** to edit the Kubernetes Job resource name, autogenerated from the display name. This must be a valid Kubernetes name.
7. In the **Project** field, select the Kubernetes namespace where the transfer job runs.

**NOTE**

If you are a non-admin user, you see only projects and namespaces to which you have access. The UI validates that the selected namespace has access to the model registry instance. If access cannot be confirmed, this displays a warning, and you must select a different project.

8. In the **Model origin location** section, click the **Object storage** or **URI** source location type.
9. Provide the details for your selected model source location:
 - For **Object storage** (S3-compatible), enter:
 - **Endpoint:** S3 or MinIO endpoint URL, for example: **https://s3.amazonaws.com**.
 - **Bucket:** Bucket name, for example: **ods-ci-s3**,
 - **Region:** AWS region, if applicable, for example: **us-east-1**.
 - **Path:** Prefix or path to the model files in the bucket, for example: **/kserve-openvino-test/openvino-example-model**.
 - **Access Key ID** and **Secret Access Key:** Credentials for the source bucket.
 - For **URI**, provide a direct URL. The supported schemas are:
 - **https://** or **http://**
 - **hf://** for HuggingFace Hub, which uses **snapshot_download**

**NOTE**

For catalog models, the URI is prefilled from the catalog source and cannot be changed.

10. In the **Model destination location** section, provide the OCI registry details:
 - **Registry:** OCI registry host such as **quay.io** or your self-hosted registry.
 - **URI:** Image reference without a scheme, for example, **myorg/granite-7b:v1**.
 - **Username:** OCI registry username.
 - **Password:** OCI registry token or password.
In the background, OpenShift AI creates a ModelCar-compatible OCI image:
 - A **busybox:latest** base image is pulled. For Quay destinations, **quay.io/quay/busybox:latest** is used automatically.
 - Model files are layered on top using **olot** (OCI Layers On Top).
 - The resulting image is pushed to your specified OCI destination.
11. Complete the remaining model metadata fields: model name, version, format, description, and so on.

12. Click **Register**.

This creates the transfer job and Kubernetes resources in the selected namespace:

- ConfigMap with model metadata.
- Secret with source credentials (S3 only).
- Secret with destination registry credentials in **.dockerconfigjson** format.
- Kubernetes Job running the async-upload container image.
The ConfigMap and Secrets are owned by the Job and are garbage-collected when the Job is deleted.

The Job container downloads from the source, layers files into a ModelCar OCI image, pushes to the destination, and calls the Model Registry API to create the model artifact with the **oci://** URI.

Verification

- A **Model transfer job started** notification message displays immediately and you can click the link to the model transfer job.
- When the transfer job completes, click the link in the relevant notification message:
 - Success: View the newly registered model and version.
 - Failure: View Kubernetes events for the job pods, and retry or delete the job.
- On success, on the newly registered model version page:
 - The **Storage location** section shows the OCI destination details.
 - The artifact URI is set to **oci://<registry>/<image>:<tag>**, which KServe ModelCar can then use for model serving.

2.4. MONITORING MODEL TRANSFER JOBS

You can monitor the model transfer jobs that are created when you store model artifacts at registration time. Model transfer jobs are background Kubernetes Jobs that copy models from a source location to an OCI registry as ModelCar images.

Prerequisites

- You have logged in to the Red Hat OpenShift AI dashboard.
- You have access to a model registry instance.
- You have created at least one model transfer job by registering and storing a model.

Procedure

1. In the OpenShift AI dashboard, click **AI hub** → **Models** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry instance in which to monitor transfer jobs.

3. On the **Models** page, click the action menu (:) beside the **Register model** button, and click **View model transfer jobs**. Alternatively, you can follow a link in a notification message. The **Model transfer jobs** table shows the following details:
 - **Job name**: Job display name and resource name.
 - **Model name** and **Model version name**: These display as links only when the transfer job is successful.
 - **Namespace**: Where the job runs.
 - **Created**: When the job was created.
 - **Author**: Username who created the job.
 - **Transfer job status**: Status indicator of **Pending**, **Running**, **Complete**, or **Failed**.
The page auto-polls while any job is in **Pending** or **Running** state.
4. Optional: In the filter drop-down list, you can select filters to refine the list of jobs, for example, by a specific job name, model name, status, or author.
5. To view detailed status and events for a job, click a status indicator in a row to open the **Status** dialog.
If the job has failed, this includes a failure alert with details.
6. In the **Status** dialog, on the **Event log** tab, you can view Kubernetes events for the job pods, for example, pulled image, started container, errors, and so on. These events auto-refresh while the job is active.

Verification

- Monitor the **Transfer job status** column for real-time updates on the progress of your transfer job.
- When a transfer job completes successfully, click the **Model name** or **Model version name** links to view your registered model version.
- If a transfer job fails, click the **Failed** status indicator to open the **Status** dialog and review the error details.

2.5. RETRYING A FAILED MODEL TRANSFER JOB

You can retry a failed model transfer job to create a new transfer job with the same configuration. Model transfer jobs are created when you choose to store model artifacts at registration time. Retrying a job creates a new transfer job and a new Kubernetes resource name.

Prerequisites

- You have logged in to the Red Hat OpenShift AI dashboard.
- You have access to a model registry instance.
- You have a failed model transfer job.

Procedure

1. From the OpenShift AI dashboard, click **AI hub → Models → Registry**.
2. From the **Model registry** drop-down list, select the model registry instance in which to retry a failed transfer job.
3. On the **Models** page, click the action menu (:) beside the **Register model** button, and click **View model transfer jobs**. Alternatively, you can also follow a link in a notification message.
4. Locate the failed job in the table, and in the **Transfer job status** column, click the **Retry** link.
5. In the **Retry model transfer job** dialog, review the prepopulated **New model transfer job name** field, and edit if necessary.
6. Optional: Click **Edit resource name** to edit the autogenerated name for the Kubernetes Job. This must be a valid Kubernetes name.
7. Optional: Ensure that **Delete the failed <job-name> transfer job** is selected to delete the job as part of the retry.
8. Click **Retry**.
The **Retry** dialog generates a new Kubernetes resource name and a new transfer job. A new job is created with the same configuration as the failed job.

Verification

- A new model transfer job displays in the **Model transfer jobs** table with a status of **Pending** or **Running**.
- If you selected to delete the failed job, the old job is removed from the table.
- Monitor the new job status to verify that the transfer completes successfully.

2.6. DELETING A MODEL TRANSFER JOB

You can delete a model transfer job to remove the Kubernetes Job and its owned resources from your namespace. Deleting a transfer job does not affect the storage location of the model if the transfer was successful.

Prerequisites

- You have logged in to the Red Hat OpenShift AI dashboard.
- You have access to a model registry instance.
- You have at least one model transfer job that you want to delete.

Procedure

1. From the OpenShift AI dashboard, click **AI hub → Models → Registry**.
2. From the **Model registry** drop-down list, select the model registry instance in which to delete a transfer job.
3. On the **Models** page, click the action menu (:) beside the **Register model** button, and click **View model transfer jobs**. Alternatively, you can also follow a link in a notification message.

4. Locate the job to delete in the table, and from the row action menu (:), click **Delete**.
5. In the **Delete model transfer job** dialog, enter the job name to confirm deletion.
6. Click **Delete**.
This deletes the transfer job and its owned resources of ConfigMap and Secrets.

Verification

- The job is removed from the **Model transfer jobs** table.
- The Kubernetes Job and its owned resources are deleted from the namespace.
- If the model was successfully transferred before deletion, the model remains available in the OCI registry and the model registry.

2.7. VIEWING REGISTERED MODELS

You can view the details of models registered in OpenShift AI, such as registered versions, deployments, and metadata associated with the model.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model that you want to view.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the model that you want to view.
3. The **Model registry** page provides a high-level view of registered models, including the model name, latest version, deployments, labels, last modified timestamp, and owner of each model. Models are sorted by their **Last modified** timestamp by default.
4. Use the search bar to find a model in the list. You can filter with a keyword by default by entering a model name, description, or label. Alternatively, click the search bar drop-down list and select **Owner** to filter by entering a model owner.
Searching by keyword performs a search across the name, description, and labels of registered models and their versions.
5. Click the name of a model to view the details page for the model:
 - a. On the **Overview** tab, you can view model metadata such as labels, description, owner, model ID, last modified and created timestamps, and custom properties, along with latest versions and deployments.
 - b. On the **Versions** tab, you can view the registered versions of the model.
 - c. On the **Deployments** tab, you can view deployments initiated from the model registry for this model.

Verification

- You can view information about the selected model on the details page for the model.

2.8. VIEWING REGISTERED MODEL VERSIONS

You can view the details of model versions that are registered in OpenShift AI, such as the version metadata and deployment information.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model version that you want to view.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the model version that you want to view.
3. Click the name of a model to view **Overview** tab on the model details page, which includes the latest model versions and deployments.
4. On the **Versions** tab, you can view the registered versions of the model.
Versions are sorted by their **Last modified** timestamp by default.
5. Use the search bar to find a version in the list. You can filter with a keyword by default by entering a model name, description, or label. Alternatively, click the search bar drop-down list and select **Author** to filter by entering a model owner.
Searching by keyword performs a search across the name, description, and labels of registered models and their versions.
6. Click the name of a version to view the version details page.
7. On the **Details** tab, you can view the **Version details** metadata, such as labels, description, custom properties, version ID, author, and last modified and registered timestamps. This also includes where the model is registered from, model location, and model format information.
You can also click **Model details** to view non-version metadata, such as labels, description, owner, model ID, last modified and created timestamps, and custom properties.
8. On the **Deployments** tab, you can view deployments initiated from the model registry for this version.
 - a. Click the name of a deployment to open its metrics page.
For information about model metrics, see [Viewing performance metrics for a deployed model](#).

Verification

- You can view the details of registered model versions on the **Model registry** page.

2.9. EDITING MODEL METADATA IN A MODEL REGISTRY

You can edit the metadata of models registered in OpenShift AI, such as the model description, labels, and custom properties. Editing model metadata affects all versions of the model.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model that you want to edit.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the model that you want to edit.
3. The **Model registry** page provides a high-level view of registered models, including the model name, latest version, deployments, labels, last modified timestamp, and owner of each model.
4. Click the name of a model to view the model details page.
5. On the **Overview** tab, you can edit metadata for the model.
 - a. In the **Labels** section, click **Edit** to edit the labels of the model, for example, **text-to-text**.
 - b. In the **Description** section, click **Edit** to edit the description of the model.
 - c. In the **Properties** section, click **Add property** to add a new property to the model, for example, **Key: license, Value: apache**.

TIP

If you enter any property value as a URL, this is displayed as a clickable link in the **Properties** section, for example: <https://www.apache.org/licenses/LICENSE-2.0>.

- i. To edit an existing property, click the action menu (⋮) beside the property, and then click **Edit**.
- ii. To delete a property, click the action menu (⋮) beside the property, and then click **Delete**.

Verification

- You can view the updated metadata on the details page for the model.

2.10. EDITING MODEL VERSION METADATA IN A MODEL REGISTRY

You can edit the metadata of model versions that are registered in OpenShift AI, such as the version's description, labels, and custom properties. Editing model version metadata affects that model version only.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model version that you want to edit.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the model version that you want to edit.
3. Click the name of a model to view the model details page.
4. Click the **Versions** tab to view the available model versions.
5. Click a version name to view the version details page.
6. In the **Version details** section, you can edit the version metadata.
 - a. In the **Labels** section, click **Edit** to edit the labels of the version, for example, **text-to-text**.
 - b. In the **Description** section, click **Edit** to edit the description of the version.
 - c. In the **Properties** section, click **Add property** to add a new property to the version, for example, **Key: license, Value: apache**.

TIP

If you enter any property value as a URL, this is displayed as a clickable link in the **Properties** section, for example: <https://www.apache.org/licenses/LICENSE-2.0>.

- i. To edit an existing property, click the action menu () beside the property, and then click **Edit**.
 - ii. To delete a property, click the action menu () beside the property, and then click **Delete**.
- d. In the **Model format** section, click **Edit** to edit the format of the model version, for example, **ONNX**.
 - e. In the **Model format version** section, click **Edit** to edit the format version of the model version.

Verification

- You can view the updated metadata on the details page for the model version.

2.11. DEPLOYING A MODEL VERSION FROM A MODEL REGISTRY

You can deploy a version of a registered model directly from a model registry.

Prerequisites

Prerequisites

- An available model registry exists in your deployment, and contains at least one registered model.
- To deploy a model version by using the model serving platform, you have fulfilled the prerequisites described in [Deploying models on the model serving platform](#).

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** list, select the model registry from which you want to deploy a model version.
3. In the **Model name** column, click the name of the model that contains the version that you want to deploy.
The details page for the model version opens.
4. Click the action menu () beside the model version that you want to deploy.
5. Click **Deploy** for the model version.
6. In the **Deploy model** dialog, configure properties for deploying the model:
 - a. From the **Project** list, select a target project.
When you select a project, OpenShift AI scans through the connections to that project and checks if any match the connections that the model was registered with.

If there is a single matching connection, this is autoselected. If there are multiple matching connections, these are displayed next in the **Connection** list.

If there are no matching connections, you can go straight to the deployment wizard, without selecting a connection, and the model location data will be autofilled as a new connection.
 - b. When there are multiple matching connections, from the **Connection** list, select the connection that you want to use when deploying the model.
The deployment wizard is autofilled with the details for your selected connection. If you selected a connection or one was autoselected, you can change the connection or create a new one in the next steps.
 - c. Click **Deploy**.
This displays the deployment wizard.
7. On the **Model details** page, provide information about the model:
 - a. From the **Model location** list, specify where your model is stored and complete the appropriate details for your connection. For a matching connection selected in the **Deploy model** dialog, the details are autofilled. Alternatively, you can create a new connection. For more information about connections, see [Adding a connection to your project](#).
 - b. Optional: In the **Model deployment name** field, enter a unique name for your model deployment. This field is autofilled with a value that contains the model name by default. This will be the name of the inference service that is created when the model is deployed.
8. Click **Next**.

9. On the **Model deployment** and **Advanced settings** pages, configure the remaining properties for deploying your model, as described in [Deploying models on the model serving platform](#).
10. On the **Review** page, review the settings that you have selected before deploying the model.
11. Click **Deploy**.

Verification

- The model deployment is displayed on the **AI hub** → **Deployments** page.
- The model deployment is displayed in the **Latest deployments** section of the model details page.
- The model version is displayed on the **Deployments** tab for the model.
- You can edit the model version deployment by clicking the action menu (⋮) beside it, and then clicking **Edit**.
- You can delete the model version deployment by clicking the action menu (⋮) beside it, and then clicking **Delete**.

2.12. EDITING THE DEPLOYMENT PROPERTIES OF A DEPLOYED MODEL VERSION FROM A MODEL REGISTRY

You can edit model version deployment properties from a model registry for models that were deployed from the registry. For example, you can change the deployment name, model framework, and source model location details.

2.12.1. Editing the deployment properties of a model deployed by using the model serving platform

You can edit the deployment properties of a deployed model version from a model registry. For example, you can change the deployment name, model framework, number of model server replicas, model server size, and source model location details.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered and deployed model version.
- You have access to the model registry that contains the model version deployment that you want to edit.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the model deployment that you want to edit.
3. In the **Model name** column, click the name of the model that contains the deployment that you want to edit.

The details page for the model opens.

4. Click the name of the model version with the deployment that you want to edit.
5. Click **Deployments**
6. Click the action menu () beside the model deployment that you want to edit.
7. Click **Edit**.
8. In the **Edit model** dialog, edit the model deployment properties:
 - a. In the **Model deployment name** field, enter a new, unique name for your model deployment.
 - b. From the **Model framework** list, select a different framework for your model.



NOTE

The **Model framework** list shows only the frameworks that are supported by the model serving runtime that you specified when you deployed your model.

- c. In the **Number of model server replicas to deploy** field, specify a value.
- d. From the **Model server size** list, select a value.
- e. In the **Model route** section, select the **Make deployed models available through an external route** checkbox to make your deployed models available to external clients.
- f. In the **Token authentication** section, select the **Require token authentication** checkbox to require token authentication for your model server. To finish configuring token authentication, perform the following actions:
 - i. In the **Service account name** field, enter a service account name for which the token will be generated. The generated token is created and displayed in the **Token secret** field when the model server is configured.
 - ii. To add an additional service account, click **Add a service account** and enter another service account name.
- g. Edit the connection by specifying an existing connection, or by creating a new connection.
- h. Customize the runtime parameters in the **Configuration parameters** section:
 - i. Modify the values in **Additional serving runtime arguments** to define how the deployed model behaves.
 - ii. Modify the values in **Additional environment variables** to define variables in the model's environment.

The **Configuration parameters** section shows predefined serving runtime parameters, if any are available.

**NOTE**

Do not modify the port or model serving runtime arguments, because they require specific values to be set. Overwriting these parameters can cause the deployment to fail.

- i. Click **Redeploy**.

Verification

- The model redeploys and is displayed with updated details on the **Deployments** tab for the model version.

2.13. DELETING A DEPLOYED MODEL VERSION FROM A MODEL REGISTRY

You can delete the deployments of model versions from a model registry.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model with a deployed model version.
- You have access to the model registry that contains the model version deployment that you want to delete.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that contains the deployment that you want to delete.
3. Click the name of a model to view more details.
The details page for the model opens.
4. Click the name of the model version with the deployment that you want to delete.
The details page for the model version opens.
5. Click **Deployments**.
6. To delete a deployment, click the action menu (⋮) beside the deployment, and then click **Delete**.
The **Delete deployed model?** dialog opens.
7. Enter the name of the model deployment in the text field to confirm that you intend to delete it.
8. Click **Delete deployed model**.

Verification

- The model deployment is no longer displayed on the **Deployments** tab for the model version.

2.14. ARCHIVING A MODEL

You can archive a model that you no longer require. The model and all of its versions will be archived and unavailable for use unless it is restored.



IMPORTANT

Models with deployed versions cannot be archived. To archive a model, you must first delete all deployments of its registered versions from the **AI hub** → **Deployments** page.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model that you want to archive.

Procedure

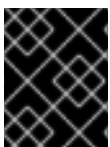
1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to archive a model in.
3. Click the action menu () beside the model that you want to archive.
4. Click **Archive model**.
5. In the **Archive model?** dialog that is displayed, enter the name of the model in the text field to confirm that you intend to archive it.
6. Click **Archive**.

Verification

- The model is no longer visible on the **Model registry** page.
- The model is displayed on the archived models page for the model registry.

2.15. ARCHIVING A MODEL VERSION

You can archive a model version that you no longer require. The model version will be archived and unavailable for use unless it is restored.



IMPORTANT

Deployed model versions cannot be archived. To archive a model version, you must first delete all deployments of the version from the **AI hub** → **Deployments** page.

Prerequisites

- You have logged in to Red Hat OpenShift AI.

- An available model registry exists in your deployment, and contains at least 1 registered model.
- You have access to the model registry that contains the model version that you want to archive.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to archive a model in.
3. In the **Model name** column, click the name of the model that contains the version that you want to archive.
The details page for the model version opens.
4. Click the action menu () beside the version that you want to archive.
5. Click **Archive model version**.
6. In the **Archive version?** dialog that opens, enter the name of the model version in the text field to confirm that you intend to archive it.
7. Click **Archive**.

Verification

- The model version is no longer visible on the details page for the model.
- The model version is displayed on the archived versions page for the model.

2.16. RESTORING A MODEL

You can restore an archived model. The model and all of its versions will be restored and returned to the registered models list.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least one archived model.
- You have access to the model registry that contains the model that you want to restore.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to restore a model in.
3. Click the action menu () beside the **Register model** button, and then click **View archived models**.
The archived models page for the model registry opens.
4. Click the action menu () beside the model that you want to restore.

5. Click **Restore model**.
6. In the **Restore model?** dialog that is displayed, click **Restore**.

Verification

- The model is displayed on the **Model registry** page.
- The model is no longer displayed on the archived models page for the model registry.

2.17. RESTORING A MODEL VERSION

You can restore an archived model version. The model version will be restored and returned to the versions list for the model.

Prerequisites

- You have logged in to Red Hat OpenShift AI.
- An available model registry exists in your deployment, and contains at least one archived model version.
- You have access to the model registry that contains the model version that you want to restore.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Registry**.
2. From the **Model registry** drop-down list, select the model registry that you want to restore a model version in.
3. In the **Model name** column, click the name of the model that contains the version that you want to restore.
The details page for the model version opens.
4. Click the action menu () beside the **Register new version** button, and then click **View archived versions**.
The archived versions page for the model opens.
5. Click the action menu () beside the version that you want to restore.
6. Click **Restore version**.
7. In the **Restore version?** dialog that opens, click **Restore**.
The details page for the version opens.

Verification

- The model version is displayed on the details page for the model.
- The model is no longer displayed on the archived versions page for the model.

